A pure head cannot attach trivially to a natural alternating tail

Human Compression project: L044 continuation

Let Ω be finite and $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$ be a string C-group. Suppose $\Omega = D \sqcup E$, $|D| \ge 2$, $|E| \ge 3$, and for some i its actual supported subgroups satisfy

$$A = G_{<i} \text{ is transitive on } D \text{ and fixes } E, \qquad B = G_{>i} = \mathrm{Alt}(E) \text{ fixes } D.$$

Put $\tau = \rho_i$, $\sigma = \rho_{i+1}$ and $H = G_{>i+1}$. Assume that for $v \in D$ and $b \in E$,

$$\tau = (v\ b)\delta, \qquad \mathrm{supp}(\delta) \subseteq E \setminus \{b\}. \tag{1}$$

In particular the crossing generator fixes every other point of the head. The requirement (1) is essential; a nontrivial head-side factor of the crossing generator is not silently dropped.

Theorem 1. *No such even string C-group exists.*

Proof. The group H fixes D and centralizes τ, so fixes b. Transitivity of $B = \langle \sigma, H \rangle$ gives $x = b^\sigma \ne b$; σ fixes D, so $x \in E \setminus \{b\}$. The involution δ is odd, since τ is even.

For each $u \in D$, transitivity of A supplies the conjugate $(u\ b)\delta$ of τ. Products of two such conjugates are the pure three-cycles on $\{b, u, w\}$, for distinct $u, w \in D$. They generate the natural $\mathrm{Alt}(D \cup \{b\})$, fixing its complement. Conjugate by σ. These two natural alternating groups have common set D of size at least two, so they generate the natural alternating group on their union. Hence

$$\mathrm{Alt}(D \cup \{b, x\}) \le P = G_{\le i+1}. \tag{2}$$

The alternating-overlap fact used here is elementary: the pure three-cycles on the two supports, conjugated using two common points, supply the three-cycles across the supports. It is valid also when either initial support has three points.

Put $Z = E \cup \{v\}$. Both B and B^τ are natural alternating groups on subsets of Z of size $|E|$ with $|E| - 1 \ge 2$ common points. Their union is Z. Since τ is even and fixes $D \setminus \{v\}$, the same overlap argument gives

$$R = G_{\ge i} = \mathrm{Alt}(Z). \tag{3}$$

Actual IP therefore gives $P \cap R = \langle \tau, \sigma \rangle$.

We first prove that τ fixes x. If $y = x^\tau \ne x$, then $y \notin D \cup \{b, x\}$, because τ preserves $D \cup \{b\}$. Conjugate (2) by τ and use overlap once more. This puts pure $\mathrm{Alt}(D \cup \{b, x, y\})$ in P. All four points v, b, x, y lie in Z, so (3) and full IP put pure $\mathrm{Alt}(\{v, b, x, y\})$ in $\langle \tau, \sigma \rangle$. A group generated by two involutions is dihedral, and all its odd-order elements lie in its cyclic rotation subgroup. But the three-cycles on $\{v, b, x\}$ and $\{v, b, y\}$ do not commute. This contradiction proves $x^\tau = x$, and hence $x^\delta = x$. This is the complete already used pure-head fixation argument.

By (2)–(3), the pure three-cycle $q = (v\ b\ x)$, or its inverse, belongs to $\langle \tau, \sigma \rangle$. Choose its orientation so that

$$\tau^q = (b\ x)\delta.$$

Write $\sigma = (b\ x)\epsilon$, where ϵ fixes $D \cup \{b, x\}$ pointwise. Both δ and ϵ are odd. Their product

$$w = \tau^q \sigma = \delta\epsilon$$

is even and supported on E, and hence belongs to the actual $B = \mathrm{Alt}(E)$. It also belongs to $\langle \tau, \sigma \rangle$. Full IP now gives

$$w \in \langle \tau, \sigma \rangle \cap B = \langle \sigma \rangle.$$

Since w fixes b and σ moves it, $w = 1$. Thus $\epsilon = \delta$ and $\sigma = (b\ x)\delta$.

The involution δ centralizes H: the latter fixes v, b and commutes with $\tau = (v\ b)\delta$. It also centralizes $\sigma = (b\ x)\delta$, because it fixes b, x. Consequently it centralizes the transitive group $B = \langle \sigma, H \rangle$ on E, while fixing $b \in E$. A centralizing permutation of a transitive action that fixes one point fixes every point. Hence $\delta = 1$, contradicting its oddness. This proves the theorem. □

Corollary 2 (The actual alternating-side subcase of source 5.31–5.32)**.** *Retain the source's supported-interval identification $A = G_{<i}$ and $B = G_{>i}$, acting on the two omission orbits D, E, respectively. Suppose A is the actual six-point head of Table 2(2) or (3), allowing full reversal of its three labels, and B is the natural alternating group on E, with $|E| \geq 3$. Suppose the crossing generator has exactly one edge between D and E. Then this configuration cannot occur in an even string C-group. The fully reversed ambient configuration is excluded as well.*

Proof. Here $i = 3$. In the displayed orientation the first two head involutions are (1 2)(3 4) and (2 3)(4 5). They generate D_{10} on $\{1, \ldots, 5\}$, with trivial centralizer on that set, and fix point 6. Their unique nontrivial orbit is preserved by ρ_3, which commutes with both. Thus ρ_3 fixes points 1 through 5. The unique crossing has endpoint 6, so $\rho_3 = (6\ b)\delta$ with $\mathrm{supp}(\delta) \subseteq E \setminus \{b\}$. All theorem hypotheses now hold.

For clarity the reversed head is not silently identified with its original ordered tuple. In reversed row 2, the first two involutions have two three-point orbits covering D, so a commuting ρ_3 preserves D: all its other orbits are singletons and cannot be exchanged with these triples. It cannot have the assumed crossing. Reversed row 3 is conjugate to its displayed orientation by (1 6)(2 5)(3 4), giving the same argument. These literal head facts are retained from the accepted ordered Table 2 and critical-head proofs; no new enumeration is used. □

Source interface and attribution. Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, physical page 26 explicitly identifies the supported sides with $G_{<i}$ and $G_{>i}$ in Section 5.2. The exceptional cases of that section are sent to Case 3 on page 29. Propositions 5.31 (page 38) and 5.32 (page 44) explicitly assume the opposite side is alternating. The corollary covers their six-point-head subcase with this actual supported identification. It does not equate an arbitrary projected alternating side with the whole actual suffix.

The accepted general critical-head reduction supplies $R = G_{\geq 3} = \mathrm{Alt}(Z)$, but does *not* by itself supply $B = G_{\geq 4} = \mathrm{Alt}(E)$. Thus that reduction alone is insufficient for this application. The stronger B assumption is explicit here; (3) then derives R independently. Neither group is identified merely from an abstract isomorphism or an orbit projection. The first pure-alternating and dihedral step is reproduced from the accepted strict-internal head lemma. The added three-cycle conjugation and full-IP extraction force the central residual involution to vanish. The theorem needs neither a rank estimate nor a finite census. Arbitrary exceptional ends without the natural B premise, remaining singleton routing, connected-square rigidity and table contracts are separate. No source-wide, formalization, question or historical-priority claim is made.